

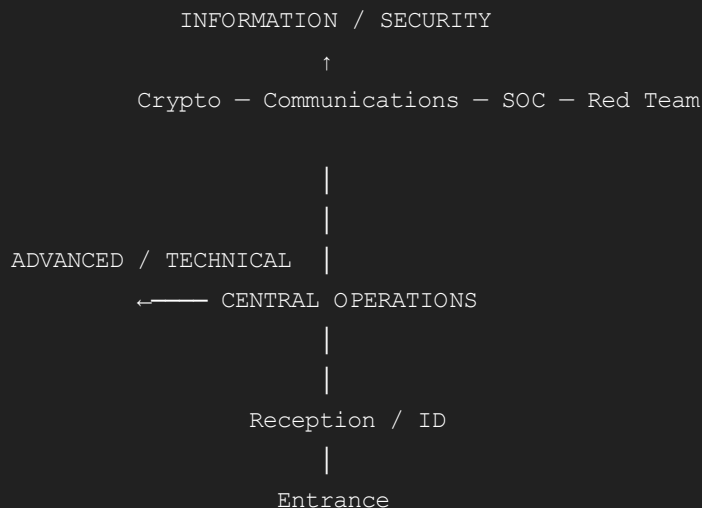
Worked for 1m 27s

Yes. **A4 is where we finally commit the A1–A3 logic to physical geometry.** We still treat this as a **canonical greybox plan v0.1**, because A5–A9 will stress-test it. But from this point onward, changes need a reason rather than preference.

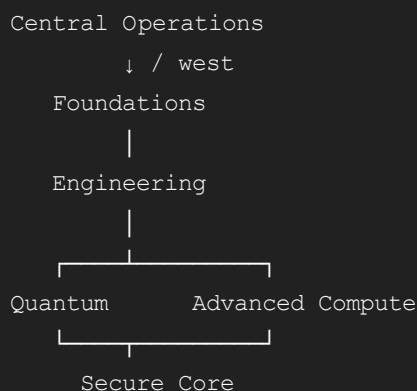
A4 — Whole Phantom Q Headquarters Floor Plan

A4.1 Design decision

I recommend a **three-branch Headquarters around Central Operations**:



The third branch extends from the western side:



The important point is that **Release 1 gameplay remains clustered around Central Ops**, while the larger engineering/quantum architecture exists from day one without forcing unnecessary walking.

A4.2 Canonical orientation

We keep the coordinate convention already established:

NORTH +Z
↑
WEST -X ← (0,0) → +X EAST
↓
SOUTH -Z

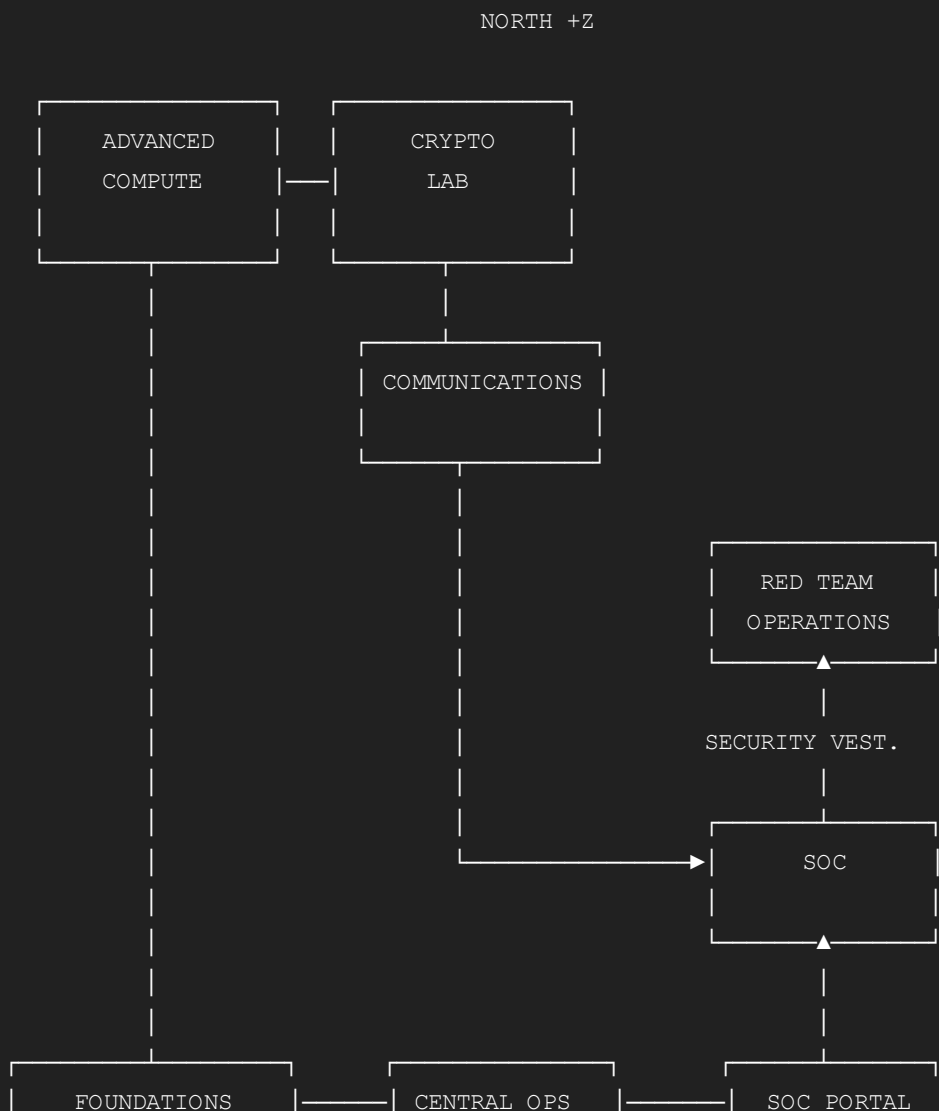
Central Operations remains:

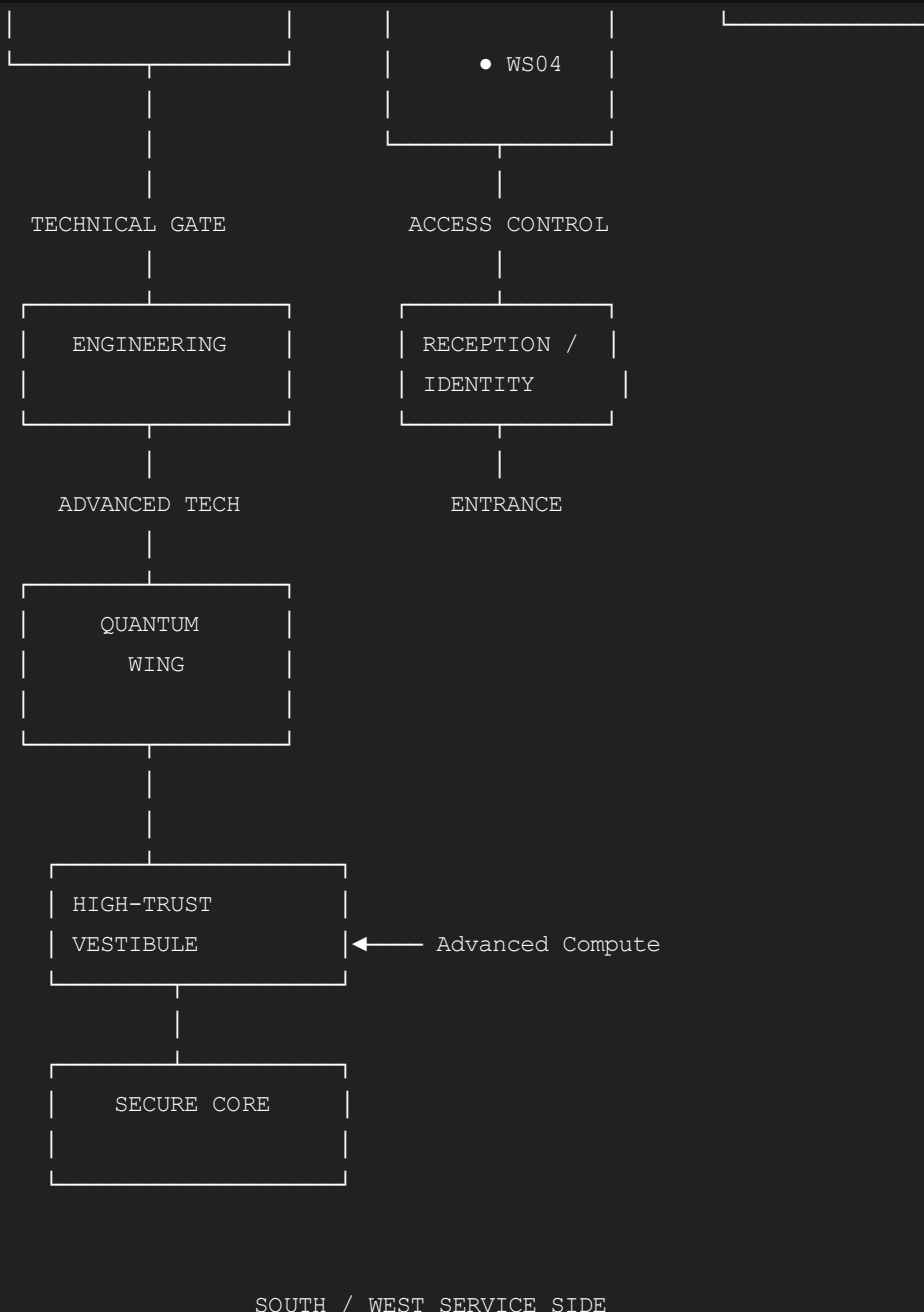
WORLD ORIGIN
X = 0
Z = 0

So the player's mental and technical reference point are the same.

A4.3 Proposed complete top-down plan

This is the **relationship-correct physical layout** I would take into Unity greyboxing.





The actual coordinate placement below makes this somewhat more compact than the diagram appears.

A4.4 Room geometry

These are the proposed **greybox dimensions and centres**.

Area	Size	Centre X	Centre Z
Central Operations	18 × 14 m	0	0
Reception / Identity	12 × 8 m	0	-17
Foundations	14 × 12 m	-18	0

Area	Size	Centre X	Centre Z
Communications	14 × 10 m	0	+17
Cryptography Lab	14 × 12 m	-18	+17
SOC	16 × 12 m	+20	+9
Red Team	16 × 12 m	+39	+9
Engineering	16 × 14 m	-19	-15
Quantum Wing	18 × 14 m	-39	-15
Advanced Compute	14 × 12 m	-35	+12
Secure Core	12 × 10 m	-53	0

These are deliberately clean metre dimensions so they snap properly to our Unity architectural grid.

A4.5 Central Operations remains the physical heart

Bounds:

CENTRAL OPERATIONS

X = -9 → +9

Z = -7 → +7

This gives us:

NORTH

COMMUNICATIONS

↑

FOUNDATIONS ← CENTRAL OPS → SOC

↓

RECEPTION / ID

So the four major directions from the room already communicate function.

That is excellent for navigation.

A4.6 Arrival sequence

South side:

```
EXTERIOR
  ↓
MAIN ENTRANCE
  ↓
RECEPTION
  ↓
IDENTITY / REGISTRATION
  ↓
T0 — HQ ACCESS CONTROL
  ↓
CENTRAL OPERATIONS
```

Reception occupies approximately:

```
X = -6 → +6
Z = -21 → -13
```

Between Reception and Central Ops we reserve approximately **6 m of transition space**.

This is not wasted corridor.

It contains:

```
registration boundary
credential station
access reader
security gate
visual reveal of Central Ops
```

So the Prologue arrival has physical staging space.

A4.7 The Central Ops reveal

I want the player to move through the access gate and have the room open visually.

Not:

```
door
↓
another narrow corridor
```

↓

Instead:

RECEPTION

↓

SECURITY GATE

↓

CENTRAL OPERATIONS

opens into full camera view

That should be one of the first visual moments of the game.

A4.8 Workstation 04 anchor

We keep the prototype location:

WORKSTATION 04

X = +4.0

Z = -2.0

That means it is:

- visible reasonably early after entering;
- not directly blocking the entrance;
- within the operational floor;
- accessible from multiple directions.

Player interaction marker:

(+4.0, 0, -3.2)

Alice marker:

(+2.8, 0, -2.4)

This remains a strong starting arrangement.

A4.9 Command and Authorisation area

Incident 06 revealed that Central Operations needs a restricted **Command / Authorisation function**.

I would put this inside the room rather than invent another department.

Approximate location:

CENTRAL OPS — NORTH-EAST INTERNAL ZONE

X $\approx$ +4 $\rightarrow$ +8

Z $\approx$ +3 $\rightarrow$ +6

Conceptually:

CENTRAL OPERATIONS

|

|— General operational floor

|— Workstations

|— Workstation 04

|— Main operational display

|— Restricted Command / Authorisation station

Later this gives the signed HQ commands a physical organisational home.

A4.10 Foundations wing

Foundations:

X = -25 $\rightarrow$ -11

Z = -6 $\rightarrow$ +6

Central Ops west edge:

X = -9

Therefore only a **2 m controlled threshold** separates them.

That is ideal.

Visually:

CENTRAL OPS

[FOUNDATIONS]

ACCESS RESTRICTED

is visible almost from the beginning.

After PQ-001:

```
ACCESS GRANTED
```

and the learner immediately has somewhere new to go.

A4.11 Communications

Communications occupies:

```
X = -7 → +7  
Z = +12 → +22
```

with a short transition from the northern side of Central Operations.

Central Ops northern wall:

```
Z = +7
```

So we have approximately **5 m** between the operational floor and Communications.

This transition can contain:

```
session-status displays  
network-route visualisation  
communications access control
```

instead of being an empty corridor.

A4.12 Crypto beside Communications

Crypto:

```
X = -25 → -11  
Z = +11 → +23
```

Communications western wall:

```
X = -7
```

This leaves a **4 m specialist connection**.

So:

CRYPTO

↔

COMMUNICATIONS

is extremely easy.

But Crypto still feels like its own controlled laboratory.

A4.13 Foundations and Crypto are close but deliberately not directly merged

Foundations north boundary:

$Z = +6$

Crypto south:

$Z = +11$

Approximately:

5 m

between them.

That allows a short specialist corridor.

Why not one shared door?

Because they represent different functions:

FOUNDATIONS

reasoning / training

CRYPTO

technical security capability

Close is appropriate.

Merged is not.

A4.14 SOC position

SOC:

```
X = +12 → +28  
Z = +3 → +15
```

This placement is particularly strong because it allows SOC to interface with **both** Central Operations and Communications.

Central Ops:

```
X ends +9
```

SOC:

```
X starts +12
```

Only about a **3 m secure transition**.

SOC ↔ Communications

Communications:

```
X ends +7  
Z = +12 → +22
```

SOC:

```
X starts +12  
Z = +3 → +15
```

Their northern/southern geometry overlaps around:

```
Z = +12 → +15
```

So we can create a second secure SOC access route:

```
COMMUNICATIONS  
  ↓  
SOC PORTAL  
  ↓  
SOC
```

This is perfect for Incident 04.

A4.15 SOC architecture should expose but protect

The Central Ops-facing side could use substantial secure glazing.

So early players see:

```
SOC OPERATORS
large evidence displays
incident boards
network activity
```

but cannot enter.

That makes SOC feel important long before Incident 04.

A4.16 Red Team

Red Team:

```
X = +31 → +47
Z = +3 → +15
```

SOC ends:

```
X = +28
```

That leaves a clean **3 m security transition zone**.

Instead of corridor:

```
SOC
↓
DOOR
↓
RED TEAM
```

we build:

```
SOC
↓
T2 RED TEAM VESTIBULE
```

↓
second access control
↓
RED TEAM OPERATIONS

This is exactly where the Incident 05:

RED TEAM — STANDBY

state can be shown.

A4.17 Security branch

The resulting east side becomes very clean:

CENTRAL OPERATIONS
↓
SOC
↓
RED TEAM

with Communications feeding SOC from above.

Conceptually:

COMMUNICATIONS
|
▼
OPS ——— SOC —[SECURE]— RED TEAM

That is probably the strongest portion of the plan.

A4.18 Engineering branch

Engineering:

X = -27 → -11
Z = -22 → -8

This puts it southwest of Foundations and Central Operations.

Why here?

It achieves three things.

It remains near **Central Ops**.

It remains near **Foundations**.

And it gives us a natural path deeper west towards Quantum without crossing public circulation.

A4.19 Technical gate

The small gap between the operational/learning area and Engineering becomes:

T3 — Technical Gate

This is where the building starts looking more industrial.

Normal operational flooring/walls begin transitioning into:

- more visible services;
- equipment signage;
- technical access control;
- heavier doors;
- hardware staging.

This makes the player feel:

I am leaving office operations and entering engineering space.

A4.20 Quantum Wing

Quantum:

X = -48 → -30

Z = -22 → -8

Engineering:

X ends = -27

So we leave approximately:

3 m

for the Engineering ↔ Quantum controlled transition.

This is ideal for:

```
personnel access
+
equipment access
```

A4.21 Quantum room gets the larger footprint for a reason

Its 18×14 m footprint is intentional.

It eventually needs to support:

```
Alice station
optical path
Eve position
Bob station
control electronics
physical QKD apparatus
learner circulation
instructor visibility
digital QKD display
```

So we are protecting that future requirement now rather than trying to squeeze hardware into a small room later.

A4.22 Advanced Compute

Advanced Compute:

```
X = -42 → -28
Z = +6 → +18
```

This location is quite useful.

It sits:

- deeper than ordinary operations;
- near Crypto;
- within the advanced technical side;
- reachable from Engineering through a controlled technical spine.

So it can later support multiple domains rather than being treated only as a Quantum sub-room.

A4.23 Advanced Compute ↔ Crypto

Crypto west boundary:

$x = -25$

Advanced Compute east boundary:

$x = -28$

Only approximately:

3 m

between them.

We can therefore have a controlled **technical connection**, but not an unrestricted shared door.

That gives future advanced-crypto gameplay an obvious location.

A4.24 Secure Core

Secure Core becomes the western/deepest terminal:

$x = -59 \rightarrow -47$

$z = -5 \rightarrow +5$

Crucially:

It does not lie on any route required to reach another department.

You only go there because you are authorised to go there.

That satisfies A2/A3.

A4.25 High-Trust Vestibule

Between Secure Core and the advanced rooms we reserve a dedicated secure transition roughly around:

X = -47 → -42

Z = -5 → +5

This becomes:

T5 — High-Trust Vestibule

It can eventually include:

```
multi-factor access
dual-door logic
identity verification
clearance verification
mission authorisation
security monitoring
```

We do not need all of that in Release 1.

But the architecture supports it.

A4.26 Quantum relationship to Secure Core

Quantum sits south of this high-trust area.

A short restricted technical link connects the northern side of Quantum into the high-trust vestibule.

So:

```
QUANTUM
  ↘
HIGH-TRUST VESTIBULE
  ↓
SECURE CORE
```

No ordinary Quantum→Secure Core office door.

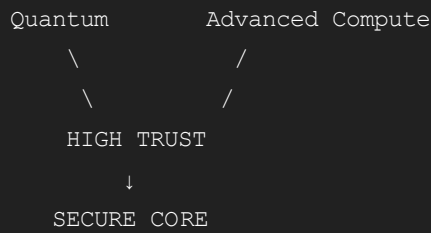
A4.27 Advanced Compute relationship to Secure Core

Likewise:

```
ADVANCED COMPUTE
  ↙
HIGH-TRUST VESTIBULE
```

↓
SECURE CORE

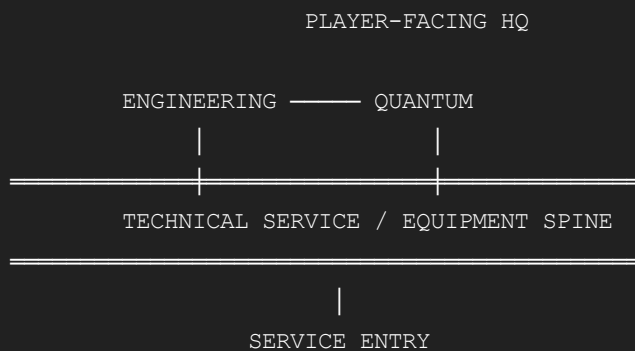
This produces exactly the hierarchy we wanted:



A4.28 Technical service spine

This is something I would now include in the canonical floor plan.

Along the southern/western side of Engineering and Quantum:



Initial width:

3 m

This route allows:

- QKD apparatus;
- replacement equipment;
- server racks;
- electronics;
- engineering tools;
- maintenance;

to enter without travelling through Reception and Central Operations.

That makes the facility much more believable.

A4.29 The service spine should initially be non-player-facing

Release 1 does not need the learner walking around loading bays.

So it can be:

```
VISIBLE SOMETIMES
+
NPC / EQUIPMENT FUNCTIONAL
+
PLAYER RESTRICTED
```

Later campaigns could use it.

Again, one building — future capability already exists.

A4.30 Approximate overall footprint

The active geometry spans approximately:

```
WEST      X ≈ -60
EAST      X ≈ +48

SOUTH     Z ≈ -26
NORTH     Z ≈ +24
```

So the entire developed bounding box is roughly:

108 m × 50 m

But this is important:

It is **not a 108 × 50 m solid rectangular building**.

The actual shape uses short wings and open/non-built gaps.

The player-facing Release 1 operational cluster is much smaller.

A4.31 Walking-distance check

With our approximate **3.2 m/s** walking speed:

Central Ops → Foundations:

~5-8 seconds

Central Ops → Communications:

~5-8 seconds

Central Ops → SOC:

~5-8 seconds

SOC → Red Team:

~4-6 seconds

Central Ops → Crypto:

~8-12 seconds

Central Ops → Engineering:

~7-10 seconds

Central Ops → Quantum:

~12-18 seconds

Central Ops → Secure Core:

~18-25 seconds

That feels right.

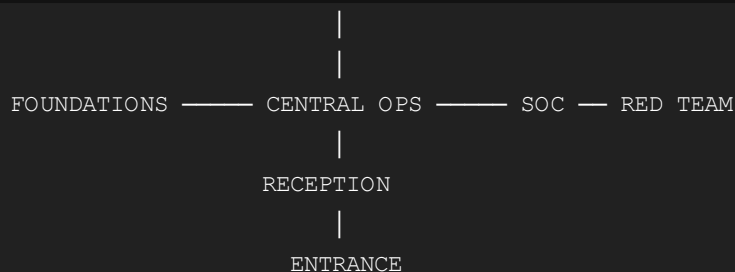
Normal operational spaces are close.

High-trust areas feel physically deeper.

A4.32 Main player circulation

The plan now has a clear hierarchy:

CRYPTO
|
COMMUNICATIONS

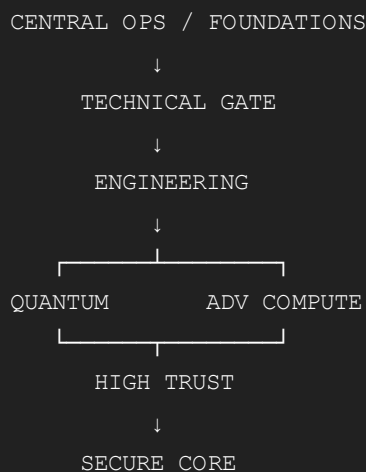


This is the **Release 1 operational loop**.

The player does not need to cross the advanced wing constantly.

A4.33 Advanced circulation

The separate technical branch gives:



This becomes increasingly important as Phantom Q expands beyond the first Act.

A4.34 Clearance thresholds now become physical landmarks

The plan gives us natural locations for:

Transition	Location	Meaning
T0	Reception → Operations	Enter Phantom Q
T1	Ops → specialist rooms	Capability access
T2	SOC → Red Team	Offensive/security authority

Transition	Location	Meaning
T3	Ops/Foundation area → Engineering	Technical capability
T4	Engineering → advanced technical areas	Quantum/advanced access
T5	Advanced area → Secure Core	Highest trust

This means we no longer need arbitrary invisible level gates.

The building performs that function.

A4.35 Camera test — first pass

This floor plan also works with our fixed raised-oblique camera because it avoids one enormous continuous room.

The player enters a zone:

```
CAMERA
↓
follows player
↓
upper walls cut away
↓
room becomes readable
```

Short connectors prevent the camera from constantly having to look through three rooms simultaneously.

Central Operations remains the only intentionally broad/open focal room.

That is good.

A4.36 Room visibility strategy

Not every room should expose itself equally.

We can use three visibility types:

Type	Example	Behaviour
Open visibility	Foundations	Easy to see from Core
Controlled visibility	SOC	Glass/security barrier; activity visible
Opaque/high-security	Red Team / Secure Core	Limited visual information

That means **access depth is communicated visually**, not only through text.

A4.37 Visual landmarks

For player navigation, I would make these readable from the hub:

```
WEST
Foundations identifier

NORTH
Communications structure/display

EAST
SOC secure glazing

SOUTH
Reception/access gate
```

Then the technical branch receives a distinct architectural gateway.

So the player can think:

```
left = learning/specialist
up = communications
right = security
down = exit
technical gate = engineering/advanced
```

No minimap required for ordinary movement.

A4.38 A1 loop

Does this still satisfy the building meaning?

Yes.

The learner physically experiences:

IDENTITY

↓

OPERATIONS

↓

SPECIALIST CAPABILITY

↓

SECURITY RESPONSE

and separately:

ENGINEERING

↓

ADVANCED TECHNOLOGY

↓

HIGH TRUST

So **A1 passes.**

A4.39 A2 loop

Does the security/spatial hierarchy survive?

Yes.

Central Ops remains the heart.

SOC is close but protected.

Red Team is deeper.

Engineering introduces technical depth.

Quantum/Advanced are deeper still.

Secure Core is terminal.

So **A2 passes.**

A4.40 A3 loop

Do our critical adjacencies work?

Entry ↔ Ops	YES
Ops ↔ Foundations	YES
Ops ↔ Communications	YES
Ops ↔ SOC	YES
Communications ↔ Crypto	YES
Communications ↔ SOC	YES
SOC ↔ Red Team	YES
Engineering ↔ Quantum	YES

And our deliberate separations remain:

Reception ↔ SOC	NO
Ops ↔ Red Team	NO
Ops ↔ Secure Core	NO
Foundations ↔ Red Team	NO
Red Team ↔ Quantum	NO

So **A3** passes.

A4.41 One issue A4 exposes

There is one thing we should keep an eye on during A5/A6:

Engineering access

At present, Engineering sits close to both Foundations and Central Ops, which is good.

But when we test campaign and character movement, we need to decide whether its **primary personnel entrance** should originate closer to:

Central Operations

or:

Foundations / specialist corridor

I currently lean towards the **specialist corridor between Foundations and Central Ops**, because Engineering should not look like another ordinary Central Ops office.

We don't need to change the floor plan yet.

A5/A6 will tell us.

A4.42 Second issue to test

Advanced Compute sits physically close to Crypto.

That is useful.

But we should ensure there is no accidental low-clearance shortcut:

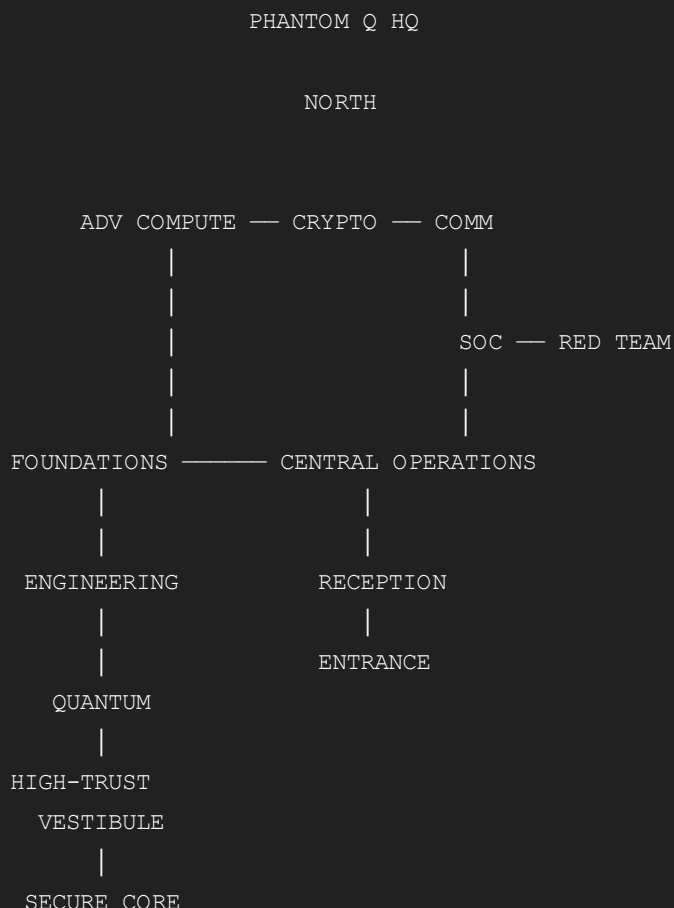
```
Crypto
↓
door
↓
Advanced Compute
```

The 3–4 m connection should therefore be a **restricted advanced-tech threshold**, not an ordinary Crypto door.

That maintains hierarchy.

A4.43 What I would now treat as our canonical A4 baseline

We now have an actual whole-building spatial model:



It gives us a **single meaningful building**, while still allowing Unity to partition and develop the building room-by-room later.

The next loop, **A5 — Campaign Routes**, should now take this exact floor plan and physically run **Prologue, Incident 01, 02, 03, 04, 05 and 06 through it scene-by-scene**. That is where we'll see whether the floor plan actually supports the narrative or merely looks architecturally coherent. If A5 exposes unnecessary walking, incorrect room use, or awkward returns to Central Ops, we correct A4 before we ever construct the full Unity greybox.